

# ***Measures of Variability /Spread***

# ***Objectives***

- Identify the measures of variability and how to use them.
- Demonstrate how to compute the measures of variability.

# *Measures of Variability*

- *Variability or dispersion* is a very important characteristic of data.
- For example, if you were manufacturing bolts, extreme variation in the bolt diameters would cause a high percentage of defective products.
- On the other hand, if you were trying to discriminate between good and poor accountants, you would have trouble if the examination always produced test grades with little variation, making discrimination very difficult.
- Measures of variability can help you create a mental picture of the spread of the data. We will present some of the more important ones.
- The simplest measure of variation is the *range*.

# ***Importance of Measures of Variability/Spread***

- *A measure of variability* is a summary statistic that represents the amount of dispersion in a dataset.
- How spread out are the values? While a measure of central tendency describes the typical value, measures of variability define how far away the data points tend to fall from the center.
- *A low dispersion* indicates that the data points tend to be clustered tightly around the center.
- *High dispersion* signifies that they tend to fall further away.

# Definition: Range

- The **range,  $R$** , of a set of  $n$  measurements is defined as the difference between the largest and smallest measurements.
- Example: The measurements in Table 1.9 vary from 5.6 to 9.4. Hence, the range is  $9.4 - 5.6 = 3.8$ .

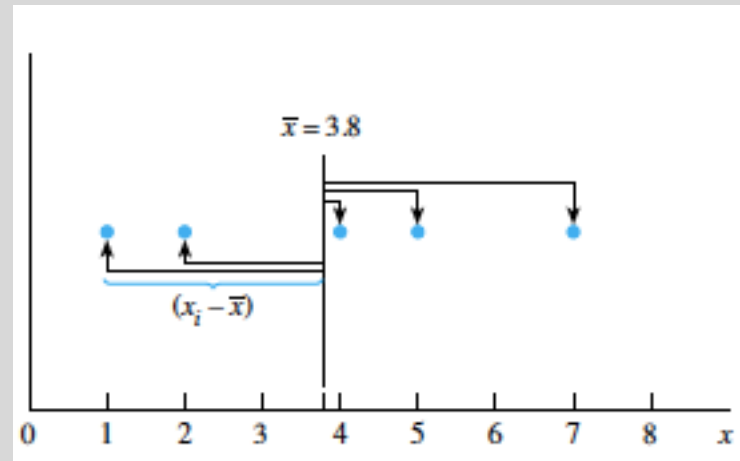
| Birth Weights of 30 Full-Term Newborn Babies |     |     |     |     |
|----------------------------------------------|-----|-----|-----|-----|
| 7.2                                          | 7.8 | 6.8 | 6.2 | 8.2 |
| 8.0                                          | 8.2 | 5.6 | 8.6 | 7.1 |
| 8.2                                          | 7.7 | 7.5 | 7.2 | 7.7 |
| 5.8                                          | 6.8 | 6.8 | 8.5 | 7.5 |
| 6.1                                          | 7.9 | 9.4 | 9.0 | 7.8 |
| 8.5                                          | 9.0 | 7.7 | 6.7 | 7.7 |

Table 1.9 Birth weights of 30 full-term Newborn Babies

# Applications of Range

- The *range* is easy to calculate, easy to interpret, and is an *adequate measure of variation for small sets of data*.
- But, for large data sets, the range is not an adequate measure of variability.
- Is there a measure of variability that is more sensitive than the range?
- Consider, as an example, the sample measurements 5, 7, 1, 2, 4, displayed as a dotplot in Figure 2.8. The mean of these five measurements is

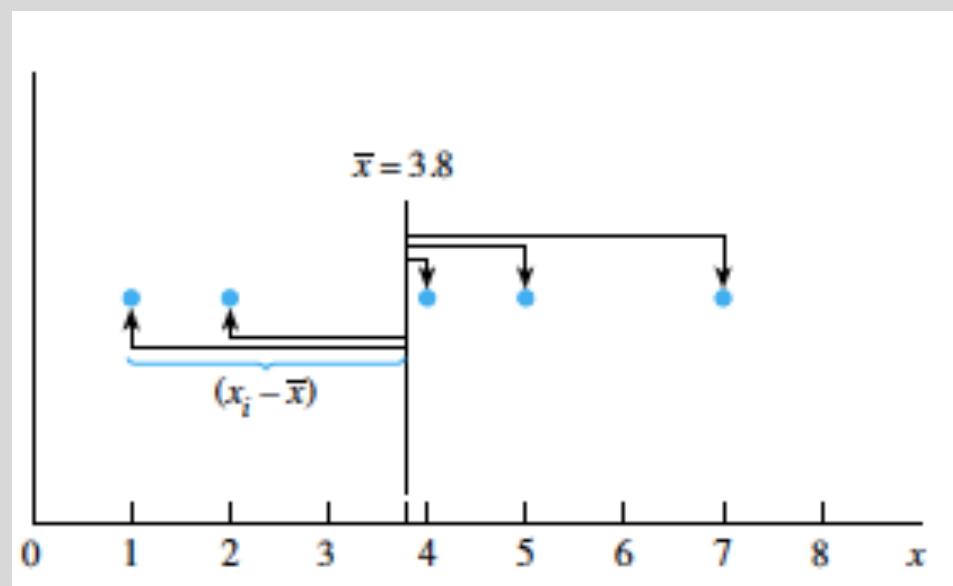
$$\bar{x} = \frac{\sum x_i}{n} = \frac{19}{5} = 3.8$$



Dotplot showing the deviations of points from the mean

# ***Applications of Range***

- The horizontal distances between each dot (measurement) and the mean  $\bar{x}$  will help you to measure the variability. If the distances are large, the data are more spread out or variable than if the distances are small.
- If  $x_i$  is a particular dot (measurement), then the deviation of that measurement from the mean is  $(x_i - \bar{x})$ .
- Measurements to the right of the mean produce positive deviations, and those to the left produce negative deviations.
- The values of  $\bar{x}$  and the deviations for our example are listed in the first and second columns of Table 2.1.



### Computation of $\Sigma(x_i - \bar{x})^2$

| $x$ | $(x_i - \bar{x})$ | $(x_i - \bar{x})^2$ |
|-----|-------------------|---------------------|
| 5   | 1.2               | 1.44                |
| 7   | 3.2               | 10.24               |
| 1   | -2.8              | 7.84                |
| 2   | -1.8              | 3.24                |
| 4   | .2                | .04                 |
| 19  | 0.0               | 22.80               |

Table 2.1



# ***Variance***

- Because the deviations in the second column of the table contain information on variability, one way to combine the five deviations into one numerical measure is to average them.
- Unfortunately, the average will not work because some of the deviations are positive, some are negative, and the sum is always zero (unless round-off errors have been introduced into the calculations).
- Another possibility might be to disregard the signs of the deviations and calculate the average of their absolute values.
- This method has been used as a measure of variability in exploratory data analysis and in the analysis of time series data.

# Variance

- We prefer to overcome the difficulty by working with their sum of squares.
- From the sum of squared deviations, a single measure called the variance is calculated.
- To distinguish between the variance of a sample and the variance of a population, we use the symbol  $s^2$  for a sample variance and  $\sigma^2$  (Greek lowercase sigma) for a population variance.
- *The variance will be relatively large for highly variable data and relatively small for less variable data.*

**Definition** The variance of a population of  $N$  measurements is the average of the squares of the deviations of the measurements about their mean  $\mu$ . The population variance is denoted by  $\sigma^2$  and is given by the formula

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

# Variance

**Definition** The **variance of a sample** of  $n$  measurements is the sum of the squared deviations of the measurements about their mean  $\bar{x}$  divided by  $(n - 1)$ . The sample variance is denoted by  $s^2$  and is given by the formula

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

The Variance is also defined as:

The average of the **squared** differences from the Mean.

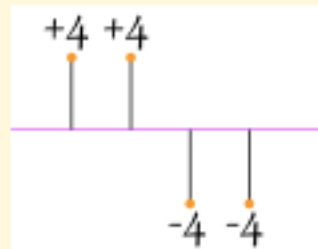
## ***How to compute for variance***

- To calculate the variance follow these steps:
- Work out the Mean (the simple average of the numbers)
- Then for each number: subtract the Mean and square the result (the *squared difference*). Then work out the average of those squared differences.

# Why square?

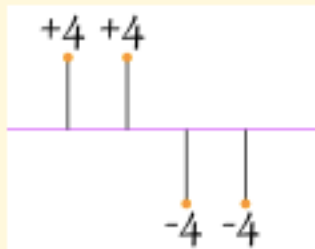
## \*Footnote: Why *square* the differences?

If we just add up the differences from the mean ... the negatives cancel the positives:



$$\frac{4 + 4 - 4 - 4}{4} = 0$$

So that won't work. How about we use absolute values?

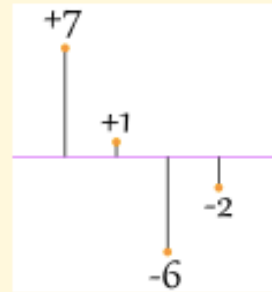


$$\frac{|4| + |4| + |-4| + |-4|}{4} = \frac{4 + 4 + 4 + 4}{4} = 4$$

# Why square?

The Standard Deviation is bigger when the differences are more spread out ... just what we want.

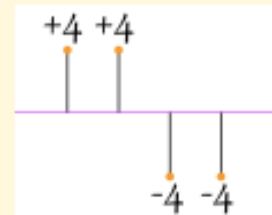
That looks good (and is the [Mean Deviation](#)), but what about this case:



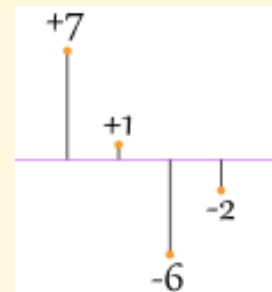
$$\frac{|7| + |1| + |-6| + |-2|}{4} = \frac{7 + 1 + 6 + 2}{4} = 4$$

Oh No! It also gives a value of 4, Even though the differences are more spread out.

So let us try squaring each difference (and taking the square root at the end):



$$\sqrt{\left(\frac{4^2 + 4^2 + 4^2 + 4^2}{4}\right)} = \sqrt{\left(\frac{64}{4}\right)} = 4$$



$$\sqrt{\left(\frac{7^2 + 1^2 + 6^2 + 2^2}{4}\right)} = \sqrt{\left(\frac{90}{4}\right)} = 4.74...$$

# Sample Variance

For the set of  $n = 5$  sample measurements presented in Table 2.1, the square of the deviation of each measurement is recorded in the third column. Adding, we obtain

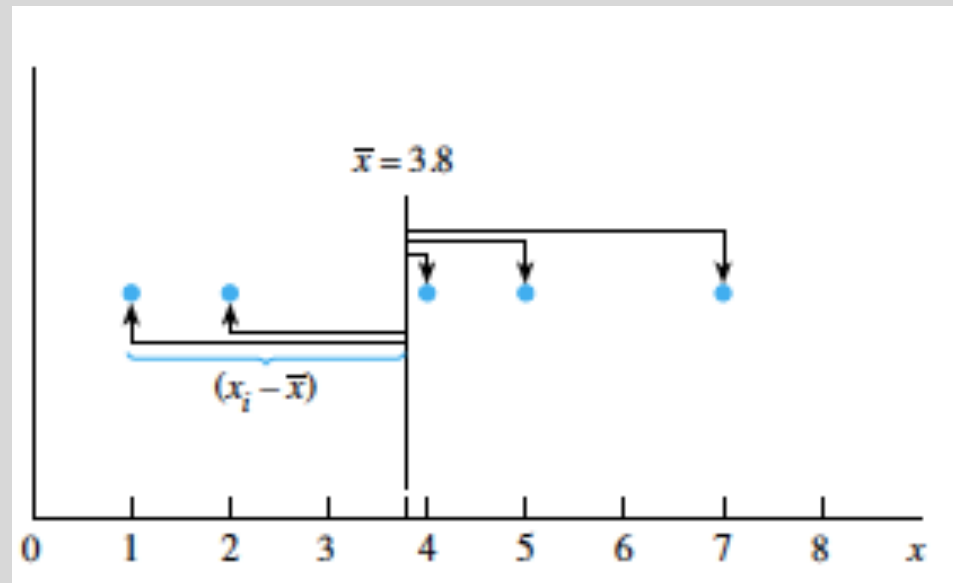
$$\Sigma(x_i - \bar{x})^2 = 22.80$$

and the sample variance is

$$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1} = \frac{22.80}{4} = 5.70$$

- The variance is measured in terms of the square of the original units of measurement.
- Taking the square root of the variance, we obtain the *standard deviation*, which returns the measure of variability to the original units of measurement.

# Sample Variance



## Computation of $\Sigma(x_i - \bar{x})^2$

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Table 2.1



# ***Standard Deviation***

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**Definition** The standard deviation of a set of measurements is equal to the positive square root of the variance.

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For the set of  $n = 5$  sample measurements in Table 2.1, the sample variance is  $s^2 = 5.70$ , so the sample standard deviation is  $s = \sqrt{s^2} = \sqrt{5.70} = 2.39$ . The more variable the data set is, the larger the value of  $s$ .

# ***Standard Deviation***

- The Standard Deviation is a measure of how spread out numbers are.
- Its symbol is  $\sigma$  (the greek letter sigma)
- The formula: it is the **square root** of the **Variance**.

# ***Properties of Standard Deviation***

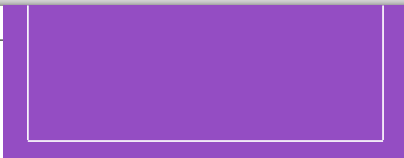
- When using standard deviation keep in mind the following properties.
- Standard deviation is only used to measure spread or dispersion around the mean of a data set.
- Standard deviation is never negative.
- Standard deviation is sensitive to outliers. A single outlier can raise the standard deviation and in turn, distort the picture of spread.
- For data with approximately the same mean, the greater the spread, the greater the standard deviation.
- If all values of a data set are the same, the standard deviation is zero (because each value is equal to the mean).

# ***Is the variance sensitive to an outlier?***

- Neither the variance, the standard deviation, nor the mean absolute deviation is robust to outliers and extreme values.
- The variance and standard deviation are especially sensitive to outliers since they are based on the squared deviations.
- A robust estimate of variability is the median absolute deviation from the median or MAD:

$$\text{Median absolute deviation} = \text{Median}(|x_1 - m|, |x_2 - m|, \dots, |x_N - m|)$$

- where  $m$  is the median. Like the median, the MAD is not influenced by extreme values.



1. Firstly find the mean:

$$\text{Mean} = (75 + 83 + 96 + 100 + 121 + 125) \div 6 = 600 \div 6 = 100$$

2. Next find the variance. To calculate the Variance, take each difference, square it, and then average the result:

$$\begin{aligned} &(75 - 100)^2 + (83 - 100)^2 + (96 - 100)^2 + (100 - 100)^2 + (121 - 100)^2 + (125 - 100)^2 \\ &= (-25)^2 + (-17)^2 + (-4)^2 + (0)^2 + (21)^2 + (25)^2 \\ &= 625 + 289 + 16 + 0 + 441 + 625 \\ &= 1,996 \end{aligned}$$

$$\text{So the Variance} = 1,996 \div 6 = 332.66\dots$$

3. The Standard Deviation is just the square root of the Variance

$$= \sqrt{(332.66\dots)}$$

$$= 18.2 \text{ correct to 1 decimal places}$$

# ***SAMPLE PROBLEM***

What is the standard deviation for the numbers:  
75, 83, 96, 100, 121 and 125?

## ***Problem 2***

- What is the variance of the first 10 numbers of the Fibonacci sequence {0, 1, 1, 2, 3, 5, 8, 13, 21, 34} ?

1. Firstly find the mean:

$$\text{Mean} = (0 + 1 + 1 + 2 + 3 + 5 + 8 + 13 + 21 + 34) \div 10 = 88 \div 10 = 8.8$$

2. Next find the variance. To calculate the Variance, take each difference, square it, and then average the result:

$$\begin{aligned} & (0 - 8.8)^2 + (1 - 8.8)^2 + (1 - 8.8)^2 + (2 - 8.8)^2 + (3 - 8.8)^2 + (5 - 8.8)^2 + (8 - 8.8)^2 + (13 - 8.8)^2 + (21 - 8.8)^2 + (34 - 8.8)^2 \\ &= (-8.8)^2 + (-7.8)^2 + (-7.8)^2 + (-6.8)^2 + (-5.8)^2 + (-3.8)^2 + (-0.8)^2 + (4.2)^2 + (12.2)^2 + (25.2)^2 \\ &= 77.44 + 60.84 + 60.84 + 46.24 + 33.64 + 14.44 + 0.64 + 17.64 + 148.84 + 635.04 \\ &= 1,095.6 \end{aligned}$$

$$\text{So the Variance} = 1,095.6 \div 10 = 109.56$$

## ***Sample Problem (Your turn)***

- The length of time required for an automobile driver to respond to a particular emergency situation was recorded for  $n = 10$  drivers. The times (in seconds) were .5, .8, 1.1, .7, .6, .9, .7, .8, .7, .8.
- Calculate the sample mean  $\bar{x}$ , variance and the standard deviation  $s$ .

Thank you!